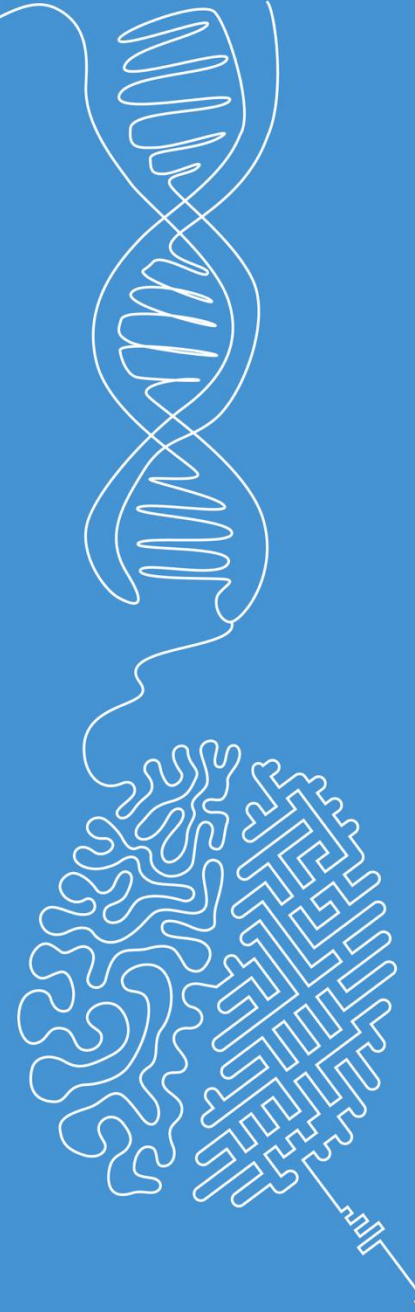


Introduction to Digital Image Processing

Part 1: Objectives and Context

Norman Juchler



Objectives

Objectives of digital image processing

- Build machines that can see
- This involves many aspects:
 - Acquire and represent image data
 - Transform and filter images
 - Enhance image data, or restore missing data
 - Optimize information content (compression)
 - Extract information from image data (sensing)
 - Understand image content (semantics)
 - Decide based on image data
 - Synthesize images

Image acquisition



Image transformation



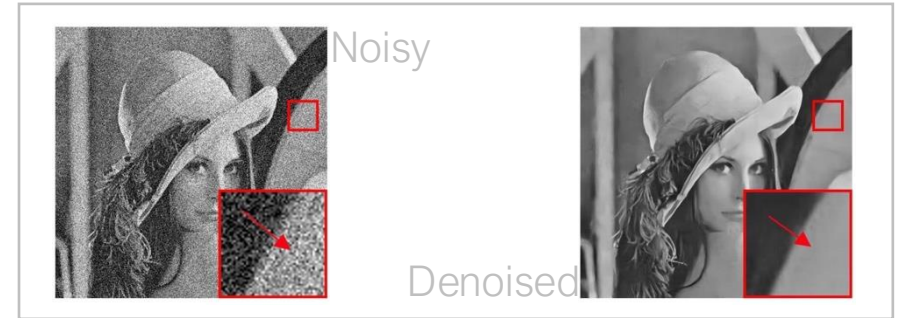
Image enhancement / restoration



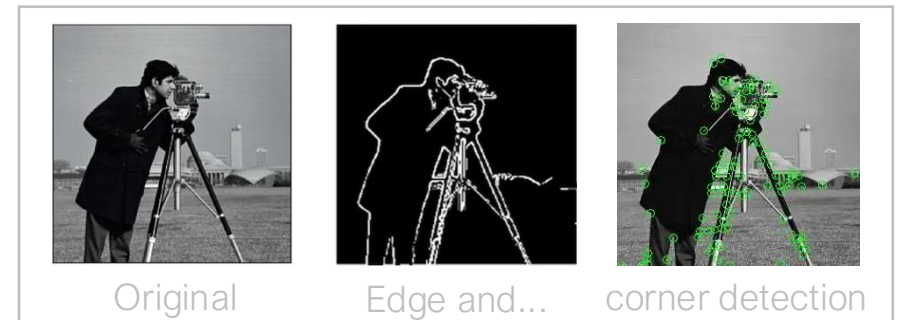
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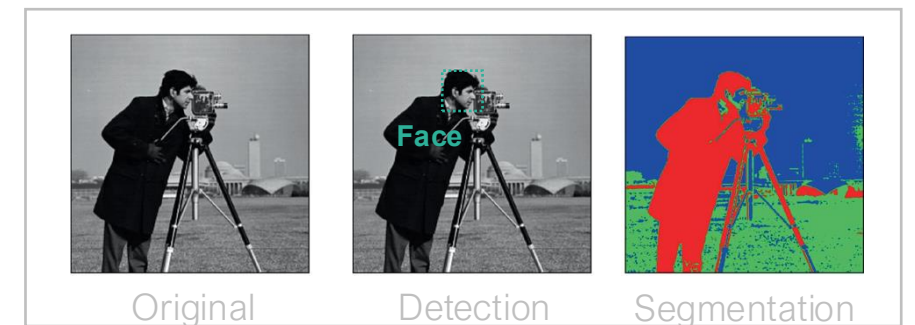
Image filtering



Extract information / features



Segmentation

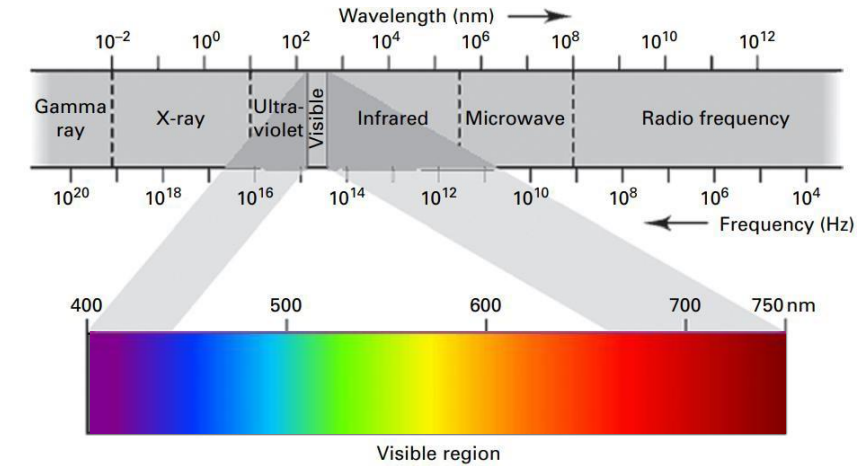


Objectives of digital image processing

- Rule of thumb: The more complex the task, the more machine learning or artificial intelligence is involved
- In this course: Image processing foundations
 - How to represent images?
 - How to process images?
 - Which are the problems to solve?
- Not in this course: Methods of machine learning / AI

Context: Imaging systems by type of radiation (or field) used

- Electromagnetic waves carry the information captured in images.
- General rules:
 - The shorter the wavelength of a wave, the better it can reveal small details
 - The short the wavelength, the more energy carries a wave
- Types of radiation or fields used in imaging:
 - Electromagnetic waves: Radio, microwaves, infrared, visible light, ultraviolet, x-rays (soft or hard), γ -rays
 - Particle radiation: Neutrons, protons, electrons, heavy ions
 - Mechanical (acoustic) waves: Ultrasound (medical imaging), sonar (underwater imaging), seismic waves (geophysical imaging)



The electromagnetic spectrum arranged according to the energy of the photons, or the frequency of the waves.

A short history of images

Or rather, a ridiculously short history

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days



Painting of a bison from the Altamira cave, dating from around 36,000 years ago. (Source: Wikimedia)

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre



1826: View from the Window at Le Gras by Niépce. Possibly the earliest surviving photograph. [Source](#)



1838: Boulevard du Temple, Paris. Photography by Louis Daguerre. [Source](#)

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre
- First digital image was taken by Russel Kirsch in 1957 using a scanner
- First digital camera using a CCD (charge-coupled device) sensor from Willard S. Boyle and George E. Smith in 1969



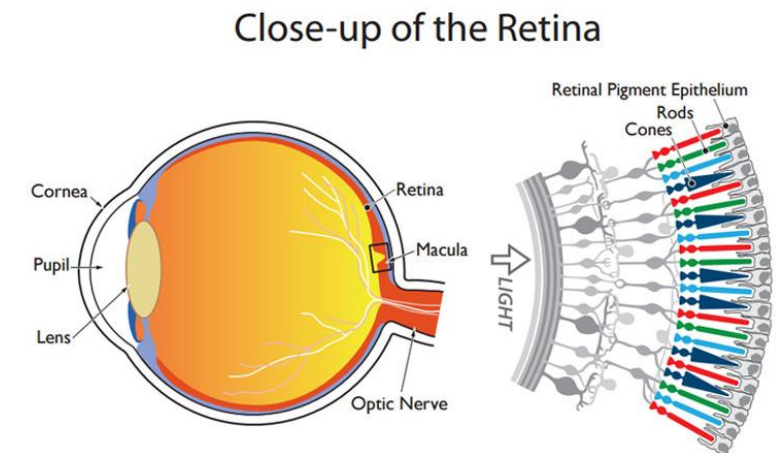
1957: First digital image ever created, by Russel Kirsch. [Source](#)

Human visual system

Take inspiration from nature

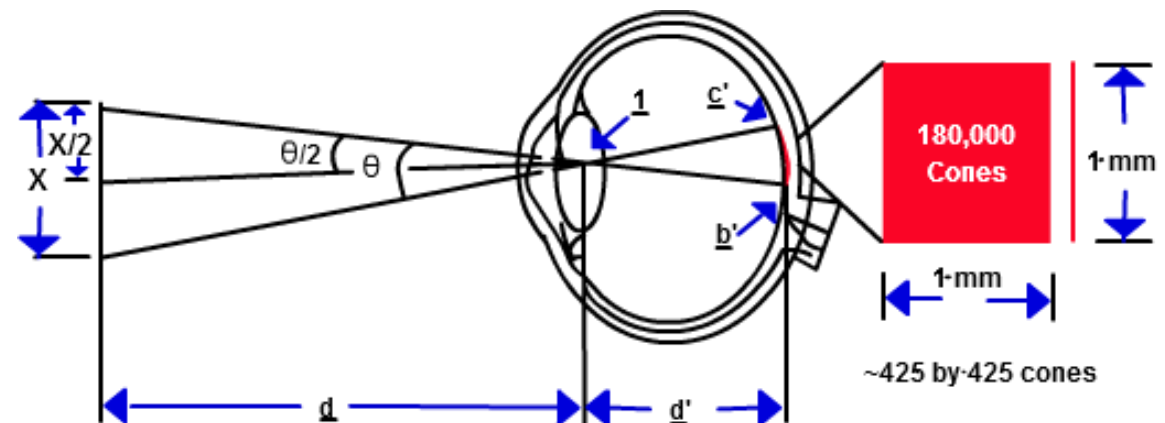
Human visual perception

- Up to about 60% of the brain activity are involved in processing visual stimuli
- More surprising facts about the human visual system
 - The image on the retina is upside-down
 - Retina is front-back inverted
 - Peripheral vision is low-res and nearly monochrome
 - The resolution of the eye is only 1 megapixel
 - The eye involves about 100 million photoreceptors
 - The eye has a special night-vision mode



Resolution (human eye)

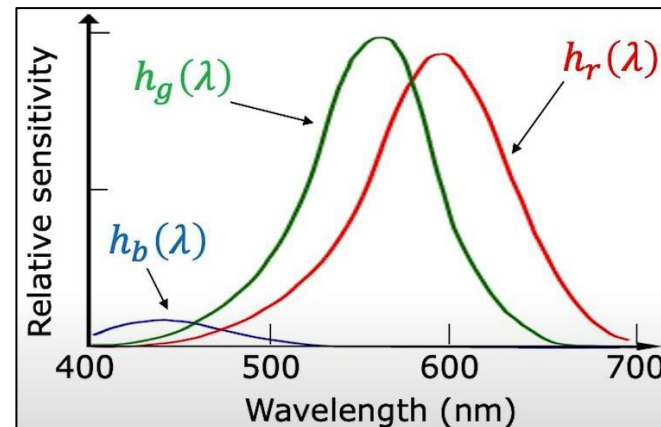
- The resolution of the human eye varies depending on several factors such as
 - the lighting conditions,
 - the contrast of the objects being observed
 - and the individual's visual acuity (Sehschärfe)
- Under optimal conditions, the human eye is estimated to have a resolution θ of around 1 arc minute
 - ...which is equivalent to $\theta = 1/60^\circ = 1/60 \cdot \pi/180$ rad
 - ...which is equivalent to about $x = 0.3\text{mm}$ for $d = 1\text{m}$
- See [here](#) for a reasoning.



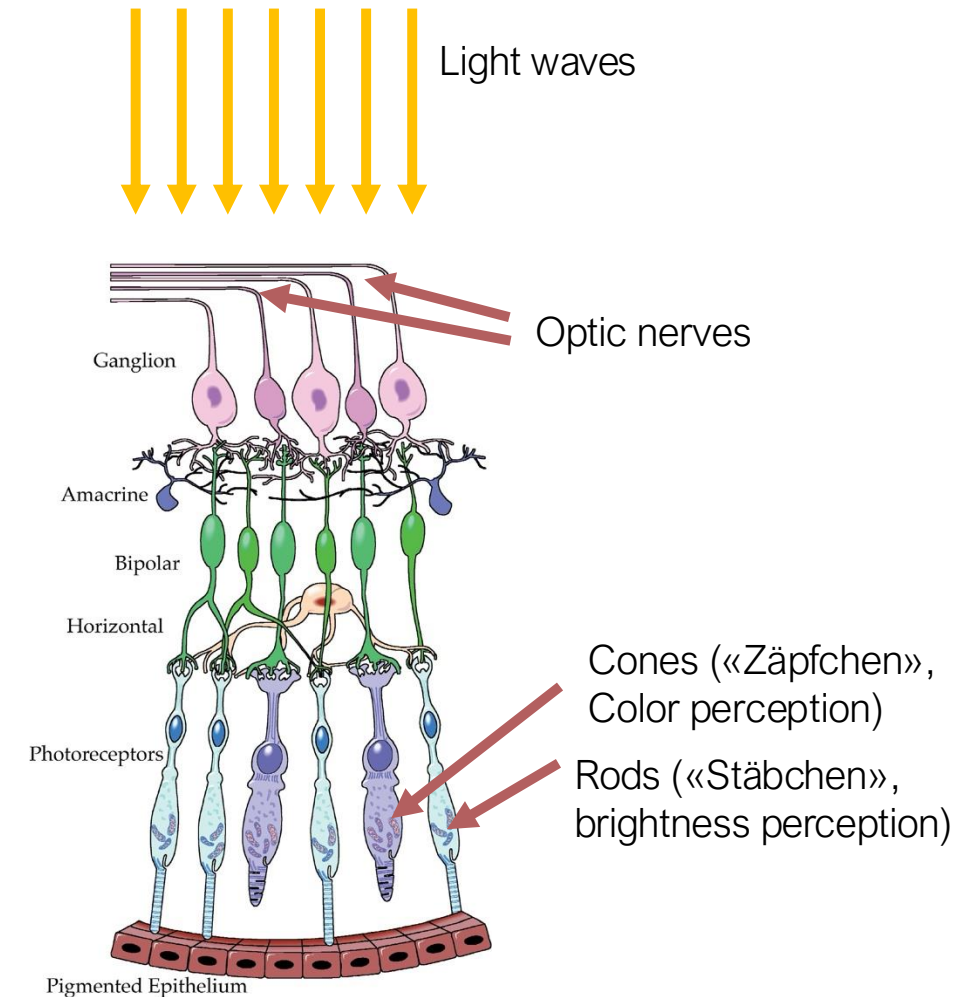
Human visual perception

- Rods and cones are specialized photoreceptor cells located in the retina of the eye.
 - Rods: very sensitive to light in general, particularly in low-light conditions, provide peripheral vision
 - Cones: Less sensitive to light, but provide the ability to perceive colors, enable high visual acuity (Sehschärfe)
- Three types of cones with different spectral sensitivity:
 - S-cones / blue
 - M-cones / green
 - L-cones / red

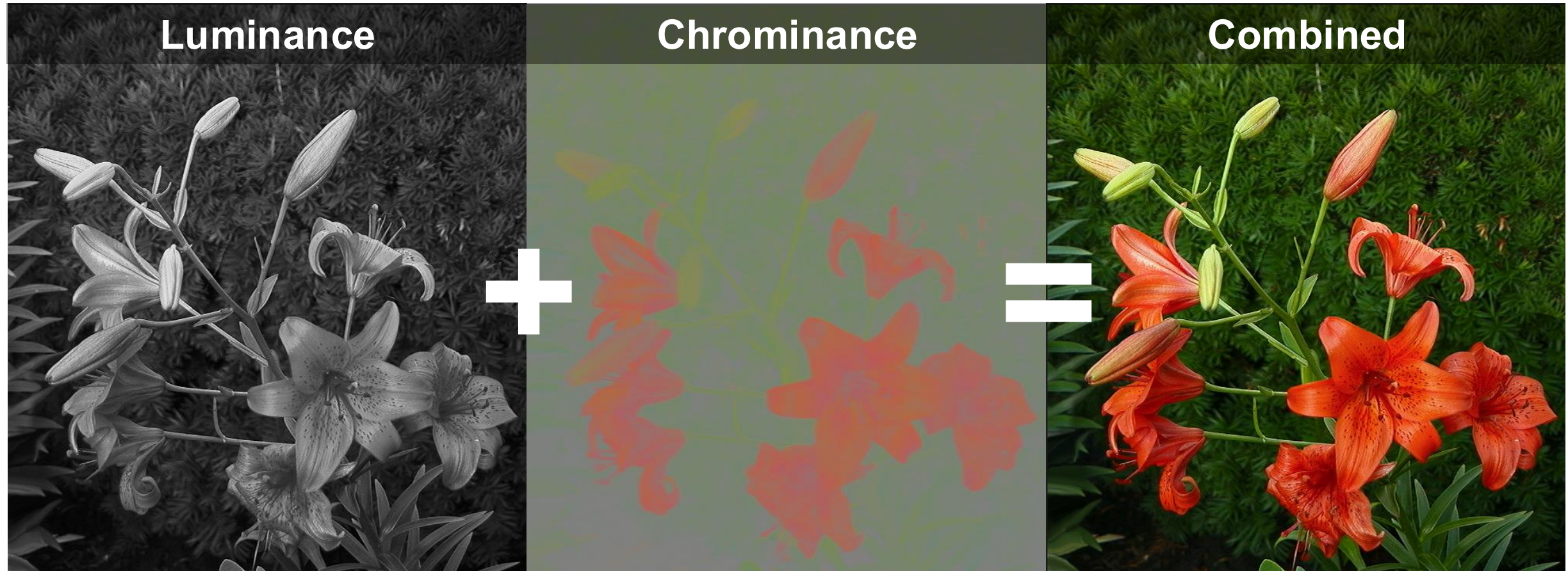
S: Short-wavelength
M: Medium-wavelength
L: Long-wavelength



«Quantum efficiency» of red, green and blue cones.



Luminance vs. Chrominance



Luminance vs. Chrominance

The human visual system processes **brightness** and **color** differently.

- **Luminance:** Brightness perception

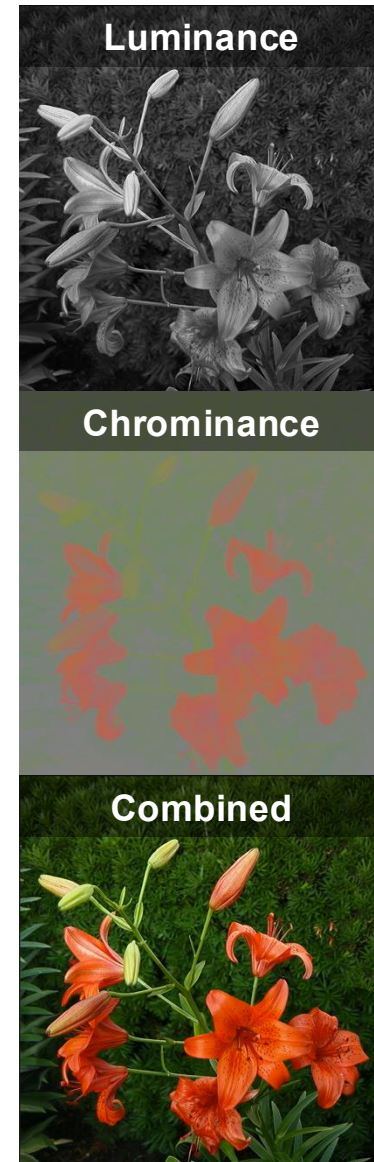
- Perceived light intensity of a scene
- Mainly processed by rods
- High sensitivity to contrast, edges, fine spatial details

⇒ Dominates our perception of **structure** and **sharpness**

- **Chrominance:** Color perception

- Represents color information (hue and saturation)
- Processed by cones
- Lower spatial resolution, less sensitive to fine detail

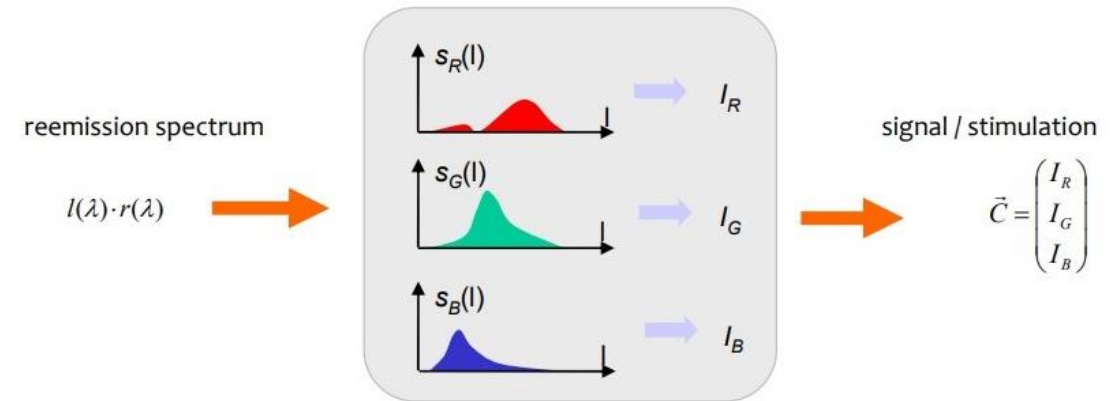
⇒ Adds **color appearance**, but contributes less to sharpness



Color perception

■ Light propagation:

- Light source spectrum: $l(\lambda)$
- Reflectance / reemission spectrum: $r(\lambda)$
- Sensitivity of cone receptors: $s_*(\lambda)$

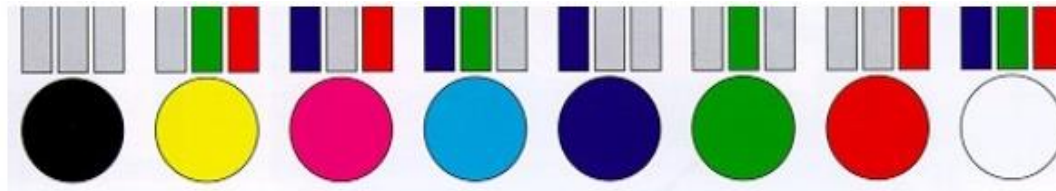


■ Result:

- Signal / stimulation consists of three channels
- Different stimulation patterns lead to the perception of different colors:

Stimuli

Perception



More on the human visual system

- Short intro to eye structure and function: [Link](#)
- Evolutionary perspective on vision, explaining some of its features: [Link](#)
- Color vision and additive color mixing: [Link](#)
- Human Vision vs. cameras – explores advanced features of our visual system: [Link](#)
- Old school explanatory video, addressing why the image is upside down and how eyesight defects occur. [Link](#)